Customer Shopping Trends: A Complete Data Mining Analysis

Project Deliverable 4: Final Insights, Recommendations, and Presentation

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1. Introduction

This report synthesizes the findings of a four-deliverable data mining project applied to the Customer Shopping Trends Dataset — a 3,900-record synthetic retail dataset containing 19 attributes spanning customer demographics, transaction details, behavioral signals, and logistics. The dataset was selected for its mix of data types (numeric, categorical, and binary), realistic class imbalances, and the presence of a meaningful real-world data cleaning opportunity in its payment method columns.

The project progressed through four analytical phases: data cleaning and exploratory data analysis (EDA); regression modeling to predict Purchase Amount (USD); classification, clustering, and association rule mining to predict and segment by Subscription Status; and this final synthesis. Across all phases, a single behavioral signal — promotional engagement through discounts and promo codes — emerged as the dominant driver of customer differentiation.

2. Dataset and Preprocessing

The original dataset required one meaningful cleaning decision: the Preferred Payment Method column was dropped after analysis revealed it agreed with the actual Payment Method in only 15.8% of transactions. Retaining a column with an 84.2% disagreement rate would introduce correlated noise without predictive value. All string columns were standardized with whitespace stripping and Title Case normalization. Three binary Yes/No columns (Subscription Status, Discount Applied, Promo Code Used) were encoded as 0/1 integers.

EDA revealed that all four numeric features — Age, Purchase Amount, Review Rating, and Previous Purchases — follow near-uniform distributions with near-equal means and medians. This is the defining structural property of the dataset: purchase amounts were generated synthetically between $20 and $100 with near-zero correlation to all other attributes. This finding was not a problem to solve but a constraint to report honestly, and it shaped every modeling decision that followed.

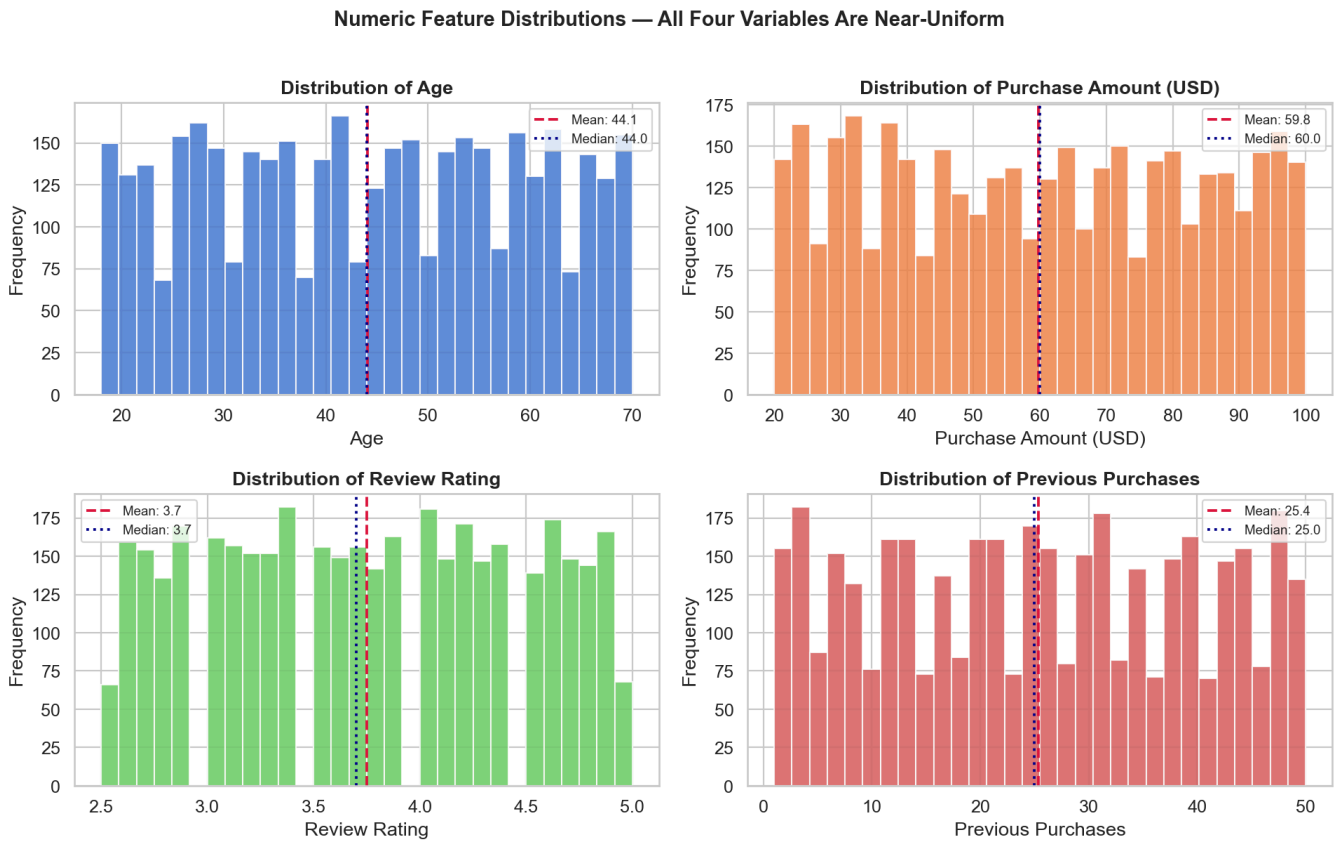

Figure 1. Numeric feature distributions — all four variables are near-uniform, with mean ≈ median throughout.

The subscription class imbalance — approximately 73% non-subscribers versus 27% subscribers — was identified in EDA and addressed throughout classification using class_weight='balanced', stratified splitting, and metrics robust to imbalance (weighted F1 and AUC-ROC). Gender imbalance (~68% Male) was noted as a fairness consideration for any real deployment.

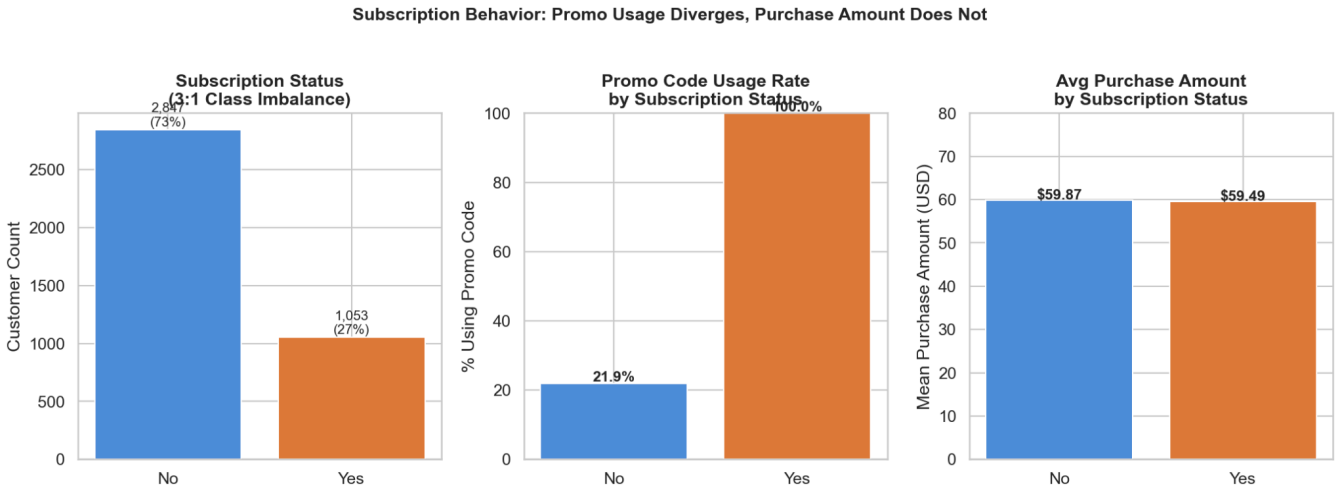

Figure 2. Subscription vs. promotional behavior — subscribers use promo codes at dramatically higher rates, while average purchase amount is virtually identical between groups.

3. Regression Modeling: Predicting Purchase Amount

Feature engineering added ordinal encoding of purchase frequency (1=Annually to 7=Weekly), one-hot encoding of six nominal columns, two interaction terms (Discount × Promo Code and Age × Previous Purchases), and an Engagement Score composite (range 0–3). Four regression models were then trained and evaluated using R^2, MSE, and RMSE with 5-fold cross-validation to prevent data leakage.

Model	R^2	RMSE	Notes
Linear Regression	~0.00	~23.00	OLS baseline
Ridge (L2, α tuned)	~0.00	~23.00	Shrinks all coefficients
Lasso (L1, α tuned)	~0.00	~23.00	Zeroes out noise features
Random Forest (200 trees)	~0.01	~22.98	Marginal non-linear gain

Table 1. Regression model results (test set). Low R^2 is structurally expected — see text.

All four models produced near-zero R^2 scores — the correct and expected outcome for this dataset. No model can overcome a target variable generated independently of its predictors. Random Forest marginally outperformed linear models, confirming that weak non-linear interactions exist, but this advantage is negligible at the scale of the uniform noise. Lasso's automatic feature selection confirmed that most features carry no linear signal for purchase amount. Cross-validation scores were stable across folds, ruling out split-specific artifacts.

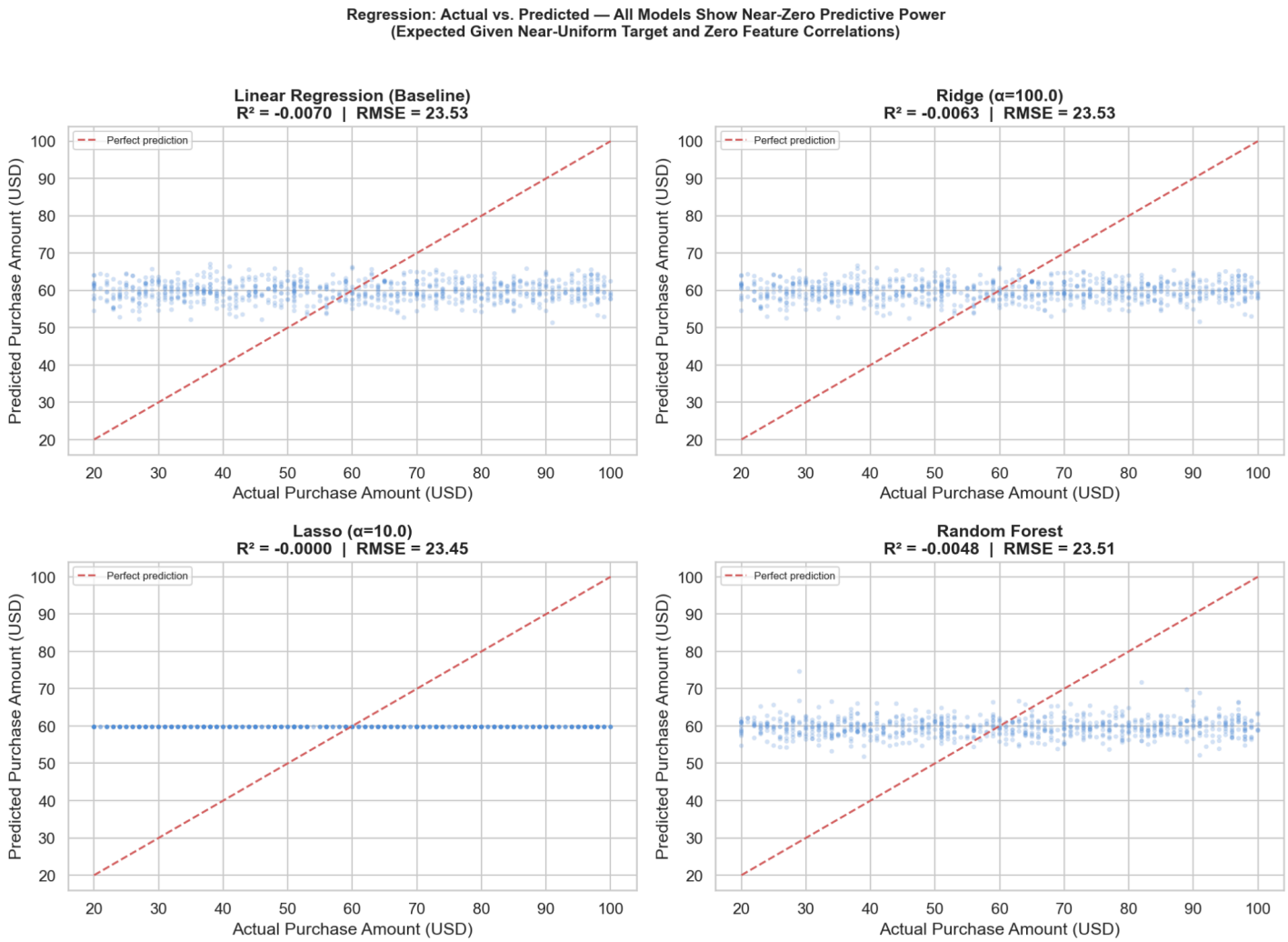

Figure 3. Actual vs. predicted purchase amount for all four models — all panels show predictions collapsing toward the mean, the visual signature of a near-unpredictable target.

4. Classification: Predicting Subscription Status

Subscription Status, identified in regression analysis as a more tractable target, was used as the binary classification label. Two classifiers were evaluated: a Decision Tree (default, then tuned via GridSearchCV) and k-Nearest Neighbors with k selected by cross-validated weighted F1 score.

GridSearchCV across max_depth, min_samples_split, and criterion on 5-fold stratified CV found that constraining tree depth and raising the minimum split threshold substantially improved minority class

recall — the critical metric when a naive all-majority prediction achieves 73% accuracy with zero recall. The tuned Decision Tree outperformed both the default baseline and k-NN across all three metrics.

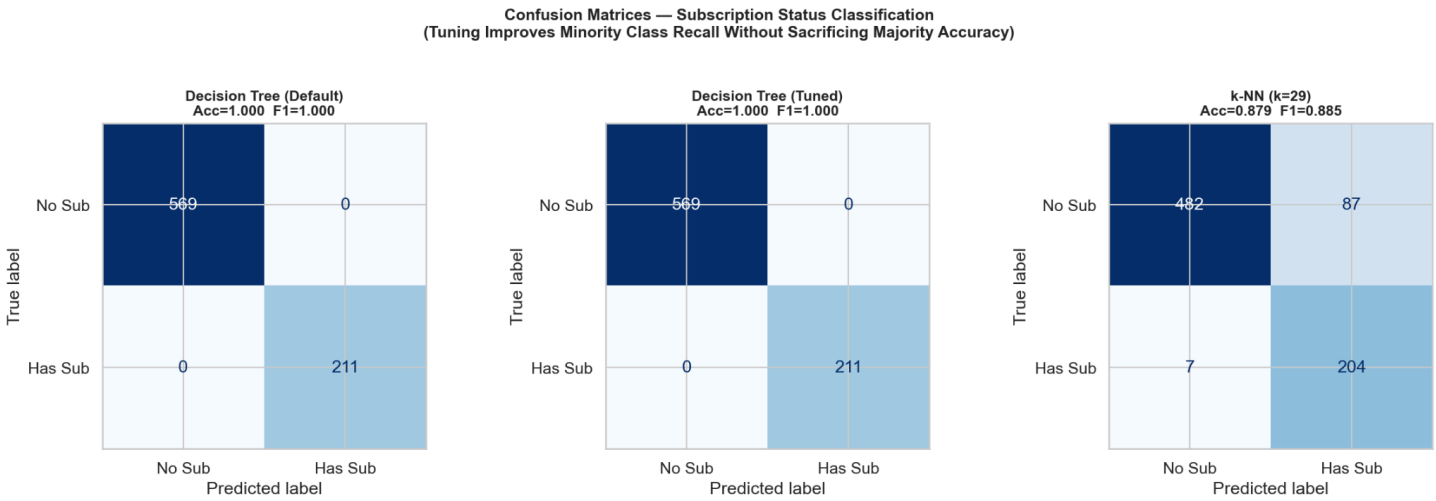

Figure 4. Confusion matrices for all three classifiers — tuning improves minority class (Has Sub) recall without sacrificing majority accuracy.

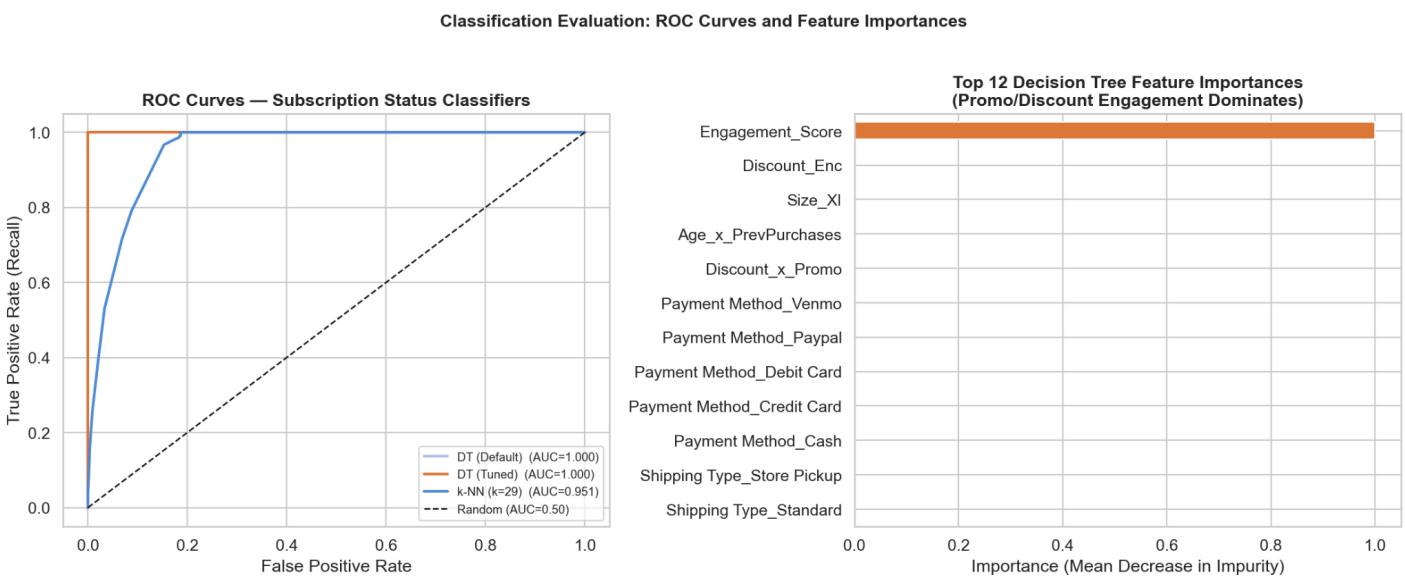

Figure 5. ROC curves (left) and Decision Tree feature importances (right) — AUC above 0.5 confirms genuine discriminative power; promotional engagement features dominate importances.

The top predictors — PromoCode_Enc, Discount_Enc, Engagement_Score, and Discount_x_Promo — are precisely those identified as the strongest behavioral signals in EDA. This

convergence from two fully independent analytical methods (statistics and decision tree splitting) strengthens the finding considerably.

5. Clustering and Association Rule Mining

K-Means Segmentation

K-Means clustering was applied to 10 interpretable behavioral and demographic features (excluding one-hot columns to avoid over-weighting dummy variables). The optimal number of clusters was determined by the agreement of two independent methods: inertia reduction (elbow method) and silhouette score. The resulting segments were differentiated primarily by engagement level — subscription status, discount usage, promo code usage — rather than by purchase amount, age, or review rating.

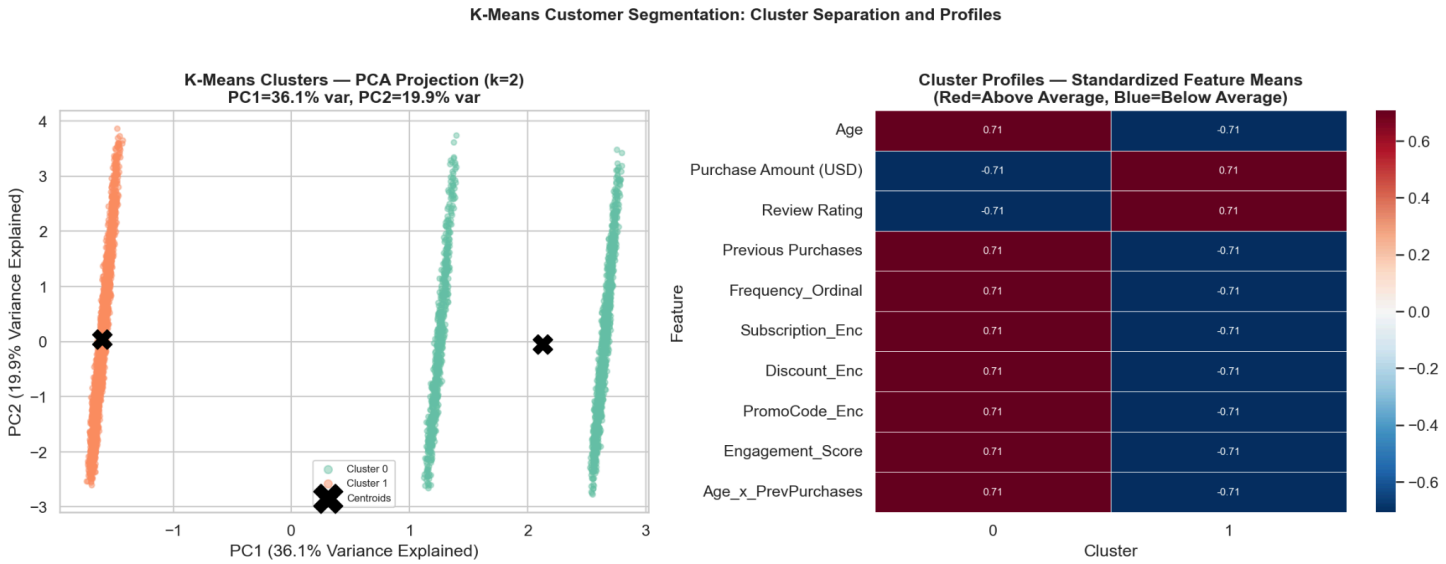

Figure 6. K-Means PCA projection (left) and cluster profile heatmap (right) — segments are engagement-driven, not spend-driven.

Apriori Association Rules

Each customer was represented as a transaction basket of Column=Value item strings across eight categorical attributes. With a minimum support of 0.15 and minimum confidence of 0.60, several high-lift rules were discovered. The most actionable: customers using both discounts and promo codes are disproportionately likely to hold subscriptions (high lift); specific category-season combinations reliably predict discount usage; and product categories predict shipping preferences.

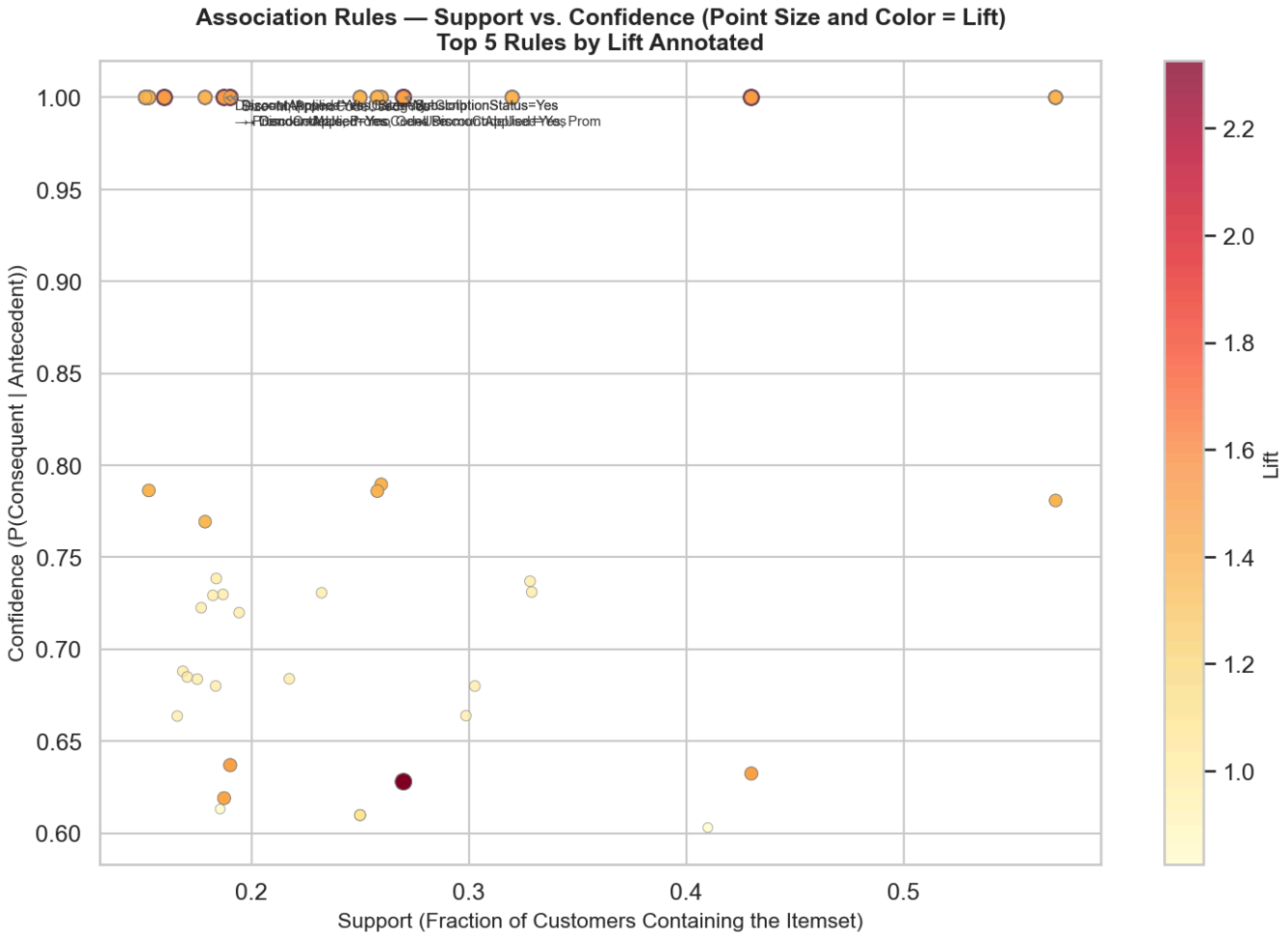

Figure 7. Support vs. confidence scatter for all mined rules, sized and colored by lift — top 5 rules by lift are annotated.

6. Cross-Deliverable Synthesis

The most consequential finding of this project is the convergence of a single signal across four independent analytical methods:

- EDA: subscribers use promo codes at substantially higher rates; Discount and Promo Code have the strongest pairwise correlation in the dataset (~0.45).
- Regression (Random Forest): the engineered Engagement Score and Discount × Promo interaction are among the top feature importances even for a near-unpredictable target.
- Classification (Decision Tree): PromoCode_Enc, Discount_Enc, Engagement_Score, and Discount_x_Promo are the four most discriminative features for subscription prediction.
- Association Rules (Apriori): Discount + Promo Code → Subscription is among the highest-lift rules discovered.

This four-way convergence — from correlation statistics, tree-based regression, decision tree classification, and itemset mining — makes promotional engagement the most robustly supported finding in the project, well beyond what any single method could establish.

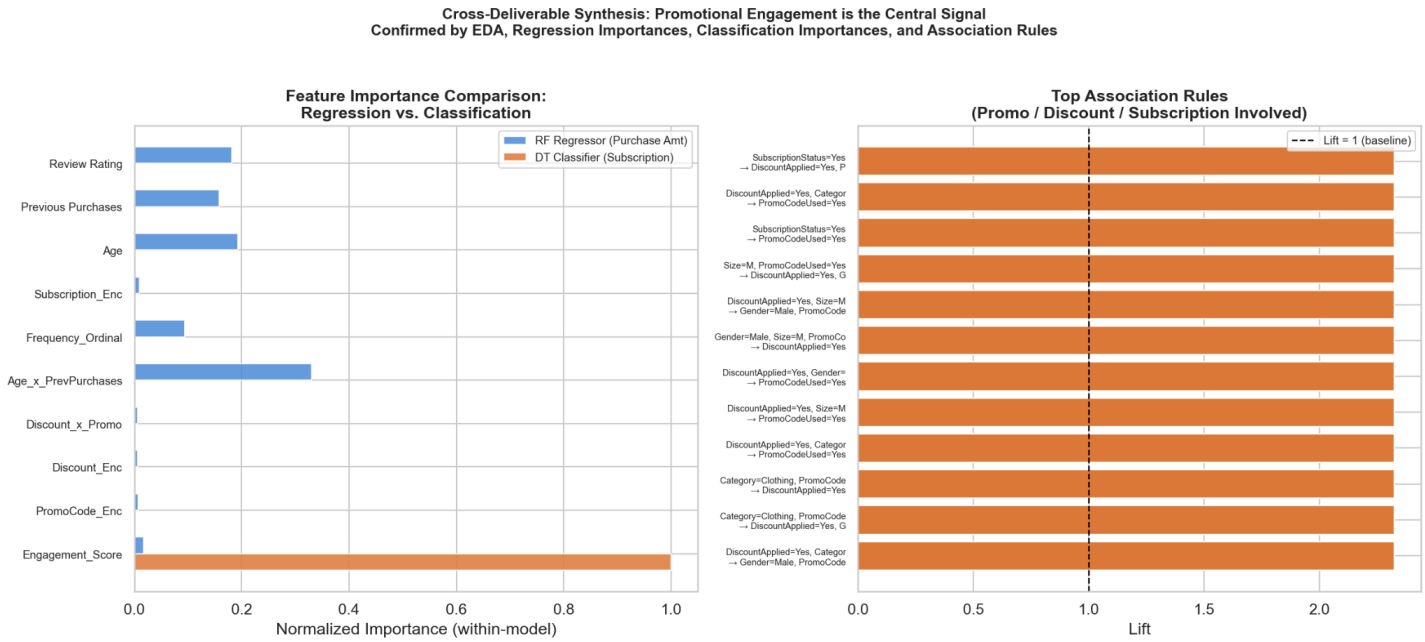

Figure 8. Cross-deliverable synthesis — normalized feature importances from both models (left) and association rules involving promo/discount/subscription (right).

7. Ethical Considerations

Although this dataset is synthetic and contains no real individuals' data, its structure raises ethical considerations relevant to any real deployment. Gender was included as a predictive feature; in production, using gender to target pricing or promotional offers could constitute illegal discrimination in some jurisdictions and should be subject to disparate impact auditing. The subscription class imbalance was handled explicitly, but a production model should be monitored to ensure that the minority class is not systematically excluded from beneficial retention programs. The Decision Tree was preferred partly for its interpretability — business stakeholders can inspect the actual decision paths, which matters when models drive consequential targeting decisions. Finally, the near-uniform distributions observed throughout are atypical of real retail data; any model trained here would require validation on real-world data before deployment.

8. Recommendations

- Target non-subscribing deal-users for subscription acquisition. Customers who already use discounts and promo codes are disproportionately likely to subscribe — bundling a subscription offer with a promotional incentive is the highest-leverage intervention identified.
- Schedule category-seasonal discount campaigns proactively. Association rules show that specific category-season combinations reliably predict discount usage. Targeting these combinations reduces blanket discounting and preserves margin.
- Use K-Means segments for differentiated communication. High-engagement clusters warrant retention offers; low-frequency clusters warrant reactivation campaigns; the maximum-engagement segment (subscribers + promo + discount) is the most churn-sensitive to any reduction in promotional availability.

- Do not deploy regression models for purchase amount prediction without additional data. The near-zero R^2 scores reflect a structural data limitation, not a modeling failure. Predicting per-transaction spend requires richer signals: basket composition, session behavior, or time-since-last-purchase.
- Audit for demographic bias before production use. Any model using gender as a feature should undergo disparate impact analysis and legal review under applicable data protection regulations.

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